Arithmetic Period Packets and the Missing Trace:

A Source-Locked Zeta Audit of Deninger's Rational-Witt Flow

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Abstract

Deninger's rational-Witt flow for $\operatorname{Spec}\mathbb{Z}$ derives an exact arithmetic clock: the periodic set is exhausted by compact packets Γ_p, indexed by rational primes, and every orbit in Γ_p has least period $\log p$. We ask whether this theorem-level period structure already defines a natural dynamical zeta function or trace. At the presently auditable level, it does not: the ordinary orbitwise product fails, while a source-intrinsic measured replacement is not yet testable. First, we prove that the orbit set in each packet is uncountable. Consequently the ordinary primitive-orbit counting measure is not locally finite in length, and the conventional orbitwise Ruelle product diverges already inside one packet for every positive real spectral parameter. This is an unconditional obstruction to the ordinary product, not an impossibility theorem for all measured or cohomological replacements. Second, we separate four interfaces: normalized Haar probability on an abstract packet base, its choice-independent lift to the source packet, relative masses across prime components, and an operator or Lefschetz trace. Only the first is presently available as an abstract compact-group theorem. On a conditional finite-support component algebra, homogeneous trace axioms leave arbitrary positive packet masses; their growth shifts the absolute-convergence abscissa from 1 to $\alpha + 1$ when $m_p \asymp p^\alpha$. Clean-family, Morse–Bott, foliated, groupoid, and microlocal trace theorems do not bypass the missing interfaces because their smooth, normal-return, measure, and operator hypotheses are not supplied by the frozen topological object. Deninger's Section 11 Haar-normalized convolution algebra is genuine but is defined on a different inverse-limit group, with no proved map to packet return traces. Zero-free deterministic controls illustrate multiplicity divergence and normalization sensitivity but are not used to infer a trace. The calibrated Route-A verdict is arithmetic origin at A0, weak packet structure at A1, no A2 pass, and no A3 or A4 test.

摘要

Deninger 为 $\operatorname{Spec}\mathbb{Z}$ 构造的有理 Witt 流给出了严格的算术时钟：周期集由素数 p 标记的紧致 packet Γ_p 穷尽，且 packet 中每条轨道的最小周期均为 $\log p$。本文检验这一周期结构是否已经足以自然定义动力学 zeta 函数或迹。结论是否定的，但否定边界是精确的。我们证明每个 packet 含有不可数条同周期本原轨道，因此通常的逐轨 Ruelle 乘积在任意正实参数处，仅在单个 packet 内就已发散。该结论只排除普通逐轨乘积，并不排除未来经额外结构定义的测度迹、群胚迹或 Lefschetz 迹。进一步地，本文区分 packet 基空间上的归一化 Haar 概率、该概率到原始 packet 的选择无关提升、不同素数分量的相对质量，以及算子迹四个独立门槛。目前只有第一个门槛能由抽象紧群定理保证。在条件性的分量代数上，齐次迹公理保留任意 packet 权重；若 $m_p \asymp p^\alpha$，绝对收敛边界随之移动到 $\Re s = \alpha + 1$。Deninger 第 11 节的 Haar 归一卷积代数属于另一逆极限群，现有来源没

有建立其与周期 packet 返回迹之间的桥梁。因此 Route A 的校准结果为：A0 具有解析算术来源，A1 较弱，A2 没有通过的候选 determinant，A3 和 A4 未进入，整体仍属探索性候选。

Keywords: arithmetic dynamics; dynamical zeta function; periodic-orbit packet; trace formula; Haar normalization; rational-Witt flow; Hilbert–Pólya programme.

1 Introduction

A periodic-orbit route to the Riemann explicit formula requires a single object to carry several logically distinct structures. Rational primes must arise intrinsically; prime powers must be repetitions; the clock must produce $r \log p$; an operator, cohomology, or transfer construction must assign the same repetitions their coefficients; and a determinant must have a controlled analytic domain and divisor. Writing the familiar prime Euler product after the primes and periods have been identified is not yet a dynamical determinant. The factor one per prime is itself a trace-normalization claim.

Deninger's arithmetic-dynamical construction makes this distinction unavoidable. For the finite-kernel admissible rational-Witt system attached to $\operatorname{Spec} \mathbb{Z}$, a rational prime does not select one isolated closed orbit. It selects a compact packet Γ_p of equal-period closed orbits [4, 5]. This is stronger than a numerical prime-orbit analogy because the prime index and period $\log p$ come from closed-point arithmetic. It is weaker than the isolated-orbit setting of a classical Ruelle product because the transverse packet multiplicity has not been converted into a trace coefficient.

The question of this paper is therefore narrow:

> Does the frozen rational-Witt flow, without importing the desired Euler product, determine a natural packet trace and dynamical determinant?

Our answer has two levels. At the ordinary orbit-counting level the answer is rigorously no: every Γ_p contains uncountably many primitive orbits of the same least period, and the conventional orbitwise product diverges. At the packetwise measured level the answer is *not yet testable*: a canonical trace domain, choice-independent lift, cross-packet normalization, and return-trace theorem have not been defined for the frozen object. The second conclusion is an interface diagnosis, not a theorem that every future enrichment must fail.

The main contributions are:

1. a source-locked reconstruction of the packet theorem, including its admissibility and choice-dependence boundaries;
2. an elementary proof that the orbit base $B_p = \widehat{\mathbb{Z}}^{\times}_{(p)}/p^{\widehat{\mathbb{Z}}}$ is uncountable;
3. a proof that the conventional individual-orbit Ruelle product cannot define a finite nonzero holomorphic product in any right half-plane;
4. a four-gate separation of local Haar probability, packet lift, global component masses, and operator trace;
5. a conditional non-uniqueness theorem showing that homogeneous trace axioms do not determine the packet masses;
6. an exact formula for how polynomial packet masses move the absolute-convergence boundary;
7. an applicability audit of clean, Morse–Bott, foliated, groupoid, and microlocal trace frameworks; and

8. deterministic zero-free controls that expose multiplicity divergence, copied-packet additivity, and mass sensitivity without claiming to construct a trace.

The paper is a Stage-2 Route-A audit. It does not inspect Riemann-zero data, fit a scale, hand-write an Euler product and relabel it dynamical, or open a general operator-algebra search. Those exclusions are methodological: they prevent the target from supplying the missing normalization.

2 Research design and source discipline

The study is a theoretical and computational mathematical case study with a comparative falsification design. The research protocol was frozen before the packet weights were calculated. It treated a rigorous nonexistence, non-uniqueness, or not-testable conclusion as a successful outcome and forbade continuation or divisor work until a trace and determinant had been legitimately defined.

The primary case is `DEN-WITT-Z-FIN`. The modular Ruelle quotient is retained only as an exact isolated-orbit coefficient benchmark; its failed rational-prime support from the preceding stage is not reopened. An equal-period circle bundle over an arbitrary compact base and a disjoint copied packet serve as proves-too-much and normalization controls.

The analysis followed five ordered steps:

1. reconstruct the source object, actions, topology claims, packets, orbit parametrization, and admissibility dependence;
2. test the conventional one-factor-per-orbit product before considering any measured collapse;
3. separate existence, local Haar normalization, packet lift, global component masses, and operator trace;
4. derive repetitions and convergence only for explicitly stipulated weights, marking them diagnostic rather than dynamical; and
5. compare the missing inputs with primary trace-formula theorems and run deterministic controls.

The preregistration imposed ten obligations. They cover an intrinsic trace domain; flow compatibility; invariance under packet automorphisms; disjoint-union additivity; transverse locality; finite-kernel compatibility; separate local and global normalization provenance; repetition covariance; an exact isolated-orbit comparison; and analytic legitimacy. Rescaling, copied packets, arbitrary compact bases, determinant direction, and finite-stability claims were adversarial controls. A design-stage Devil's Advocate audit passed the protocol subject to keeping packet-base probability distinct from packet trace.

The literature investigation used original articles, author manuscripts, and publisher or institutional copies verified through 13 August 2026. Search families targeted clean fixed sets, Morse–Bott periodic families, foliated flow traces, groupoid Haar systems, transverse measures, flat traces, and Ruelle determinants. Fifteen work-level candidates were screened; eleven core source groups and two supporting sources were retained. General operator-algebra constructions and Riemann-zero trace programs were excluded because they would change the frozen flow question. A final independent proof audit rechecked the uncountability argument, finite-subset product net, conditional mass theorem, status calibration, test run, and artifact hashes.

Non-elementary source claims are cited to their primary publications. The elementary group and series arguments are displayed in full. Computation is used only to reproduce finite consequences of those arguments, never to upgrade an open interface to a theorem.

3 Evidence boundary and frozen object

3.1 Evidence vocabulary

Four labels are used throughout:

PROVED a statement in the audited primary source or a consequence with a complete argument given here;

CONDITIONAL a theorem after explicitly listed extra assumptions, which are not all supplied by the frozen source;

OPEN a mathematically meaningful construction or property not established or refuted for the frozen object;

NOT TESTABLE an obligation that cannot yet be evaluated because its domain, topology, measure, action, or operator has not been defined.

An OPEN or NOT TESTABLE interface is not converted into a negative existence theorem. Conversely, a familiar formal expression is not counted as evidence that the missing interface exists.

3.2 Version and admissibility lock

The primary source manifestation audited here is arXiv:1807.06400v4, cross-linked by DOI to the 2026 journal article [5]; no page-perfect identity with the version of record is assumed. The author's later overview, checked as arXiv:2301.11643v1, is used only as a corroborating restatement [4]. Let $\mathfrak{X}_0 = \operatorname{Spec}\mathbb{Z}$. The source begins with a rational-Witt point space, passes to a Frobenius colimit $\check{X}_0(\mathbb{C})_{\mathcal{E}}$ carrying a $\mathbb{Q}_{>0}$-action, and forms the suspension

$$X_0(\mathcal{E}) = \check{X}_0(\mathbb{C})_{\mathcal{E}} \times_{\mathbb{Q}_{>0}} \mathbb{R}_{>0}, \qquad \phi^t[P,u] = [P, \mathrm{e}^t u]. \tag{1}$$

The source offers several admissible character classes and explicitly notes that the best choice is unclear and that topology depends strongly on it [5]. We freeze the finite-kernel class $\mathcal{E}_f$. It is source-permitted and is sufficient for the full packets in the periodic-point theorems; it is not asserted to be the unique canonical admissibility condition. Every verdict below applies to this versioned object, denoted `DEN-WITT-Z-FIN`.

3.3 What the packet theorem actually supplies

Let x_0 be a closed point with finite residue field and norm Nx_0. Deninger's Theorems 5.2 and 6.1 construct a pre-suspension packet C_{x_0} and its suspended packet

$$\Gamma_{x_0} = C_{x_0} \times_{\mathbb{Q}_{>0}} \mathbb{R}_{>0}.$$

For $\operatorname{Spec}\mathbb{Z}$, $x_0 = (p)$ and $Nx_0 = p$. The periodic set is exhausted by disjoint packets,

$$\operatorname{Per}(X_0(\mathcal{E}_f)) = \coprod_{p \text{ prime}} \Gamma_p, \tag{2}$$

and every point in Γ_p has additive flow isotropy $(\log p)\mathbb{Z}$. Thus every fibre orbit has least period $\log p$, and its r-fold return time is $r \log p$. Packet existence, disjointness, periodic exhaustion, and the arithmetic clock are PROVED.

After choosing a point above x_0 and an embedding of roots of unity, the source gives an equivariant set-level packet description. In the $\operatorname{Spec}\mathbb{Z}$ specialization it has the form

$$\Gamma_p \;\longleftrightarrow\; B_p \times \big(\mathbb{R}/(\log p)\mathbb{Z}\big), \qquad B_p = \widehat{\mathbb{Z}}_{(p)}^{\times}/p^{\widehat{\mathbb{Z}}} = \frac{\operatorname{Aut}(\overline{\mathbb{F}}_p^{\times})}{\operatorname{Aut}(\overline{\mathbb{F}}_p)}. \tag{3}$$

The circle factors are the flow orbits. Equations (37)–(39) in the source, including the fibration map to B_p, depend on the auxiliary choices. Only a different projection, equation (40), to $\mathbb{Q}_{>0}/p^{\mathbb{Z}}$ is declared canonical [5].

Accordingly, (3) is used as a noncanonical $\mathbb{R}$-equivariant bijection of sets. The source does not prove that $\Gamma_p/\mathbb{R}$ with its quotient topology is homeomorphic to B_p, or that Γ_p is a locally trivial product bundle. It does define topologies on the ambient spaces and a continuous flow, but some global equivariant continuous bijections in the construction are explicitly not homeomorphisms in general. Upgrading the displayed packet coordinates requires a separate theorem.

Table 1: Source-locked packet facts and their evidence status.

Obligation	Status	Boundary
packet existence, compactness, and exhaustion	PROVED	Theorems 5.2 and 6.1 and the author's restatement
least period $\log p$	PROVED	isotropy is $p^{\mathbb{Z}}$, or $(\log p)\mathbb{Z}$ in additive time
orbit set indexed by B_p	PROVED	noncanonical set-level parametrization
canonical topological identity $\Gamma_p/\mathbb{R} = B_p$	OPEN	quotient topology and transition maps are not identified
locally trivial packet bundle	OPEN	no local-section or properness theorem is supplied
selected packet groupoid	NOT TESTABLE	several external choices are possible; none is source-locked
packet return trace and determinant	NOT TESTABLE	no trace domain or determinant theorem is defined

4 Uncountable equal-period multiplicity

The first obstruction does not depend on the unresolved packet topology. It uses only the source's set-level orbit parametrization. Put

$$G_p = \widehat{\mathbb{Z}}_{(p)}^{\times} = \prod_{\ell \neq p} \mathbb{Z}_\ell^{\times}, \qquad H_p = p^{\widehat{\mathbb{Z}}} \subset G_p.$$

Here H_p is the closed procyclic image of the continuous exponent map $\widehat{\mathbb{Z}} \to G_p$; no injectivity assertion is needed.

Proposition 1 (Uncountable orbit base). *For every rational prime p, the compact Hausdorff quotient group $B_p = G_p/H_p$ has cardinality at least $2^{\aleph_0}$. Hence Γ_p contains uncountably many distinct primitive flow orbits, all with least period $\log p$.*

Proof. Let

$$I_p = \{\ell : \ell \text{ is an odd prime and } \ell \neq p\}.$$

This set is countably infinite, including when $p = 2$. Define

$$S_p = \left\{ (\varepsilon_\ell)_{\ell \neq p} \in G_p : \begin{array}{ll} \varepsilon_\ell \in \{\pm 1\}, & \ell \in I_p, \\ \varepsilon_\ell = 1, & \ell \notin I_p \end{array} \right\}.$$

For odd ℓ, the units 1 and -1 in $\mathbb{Z}_\ell$ are distinct. Therefore

$$S_p \cong \prod_{\ell \in I_p} C_2, \qquad |S_p| = 2^{\aleph_0},$$

and every element of S_p has order dividing two.

A procyclic profinite group has at most one nonidentity involution. Indeed, if x and y were distinct nonidentity involutions in H_p, choose an open normal subgroup U containing none of x, y, xy. The finite cyclic quotient H_p/U would then contain two distinct nonidentity involutions, which is impossible. Hence $|S_p \cap H_p| \leq 2$.

If $q : G_p \to B_p$ is the quotient map, then

$$\ker(q|_{S_p}) = S_p \cap H_p, \qquad |q(S_p)| = |S_p/(S_p \cap H_p)| = 2^{\aleph_0}.$$

Thus B_p is uncountable. Through the source-audited noncanonical set-level parametrization, the set of flow orbits in Γ_p has this cardinality. Since $\log p$ is the least period of every such orbit, each is primitive. $\square$

Remark 2. Proposition 1 is a cardinality theorem. It does not assert a homeomorphism $\Gamma_p/\mathbb{R} \cong B_p$, continuity of the orbit family, or local triviality of the source's fibration.

Corollary 3 (Failure of length-local finiteness)**.** *Let $\mathcal{P}$ be the set of individual primitive periodic orbits of the frozen flow. For every $T \geq \log 2$,*

$$\#\{\gamma \in \mathcal{P} : \ell(\gamma) \leq T\}$$

is uncountable. In particular, the ordinary primitive-orbit counting measure $\sum_{\gamma \in \mathcal{P}} \delta_{\ell(\gamma)}$ is not locally finite.

Proof. Choose any prime $p \leq \mathrm{e}^T$. Proposition 1 gives uncountably many primitive orbits at the single length $\log p \leq T$. $\square$

Theorem 4 (Orbitwise Ruelle-product obstruction)**.** *Under the conventional convention of one factor per individual primitive orbit, the formal product*

$$Z_{\text{orb}}(s) = \prod_{\gamma \in \mathcal{P}} (1 - \mathrm{e}^{-s\ell(\gamma)})^{-1} \tag{4}$$

does not define a finite nonzero ordinary Euler product on any right half-plane. For every real $\sigma > 0$, the finite-subset net already diverges to $+\infty$ within any fixed packet Γ_p.

Proof. Fix p and real $\sigma > 0$. Every primitive orbit in Γ_p contributes the same factor

$$a_{p,\sigma} = (1 - p^{-\sigma})^{-1} > 1.$$

For a finite subset $F \Subset \mathcal{P}_p$ of packet orbits, the corresponding partial product is $a_{p,\sigma}^{|F|}$. Given $M > 0$, choose n such that $a_{p,\sigma}^n > M$, and choose F_0 containing n distinct orbits. Every $F \supseteq F_0$ then satisfies

$$\prod_{\gamma \in F} (1 - p^{-\sigma})^{-1} = a_{p,\sigma}^{|F|} \geq a_{p,\sigma}^n > M.$$

The net therefore diverges to $+\infty$.

For completeness, at a complex point where the common factor $a_p(s) = (1 - p^{-s})^{-1}$ is defined, it is never equal to one. If the finite-subset net converged to a nonzero limit L, adjoining one fresh orbit would force both $Z_F(s) \to L$ and $a_p(s)Z_F(s) \to L$, hence $a_p(s)L = L$, a contradiction. At $p^{-s} = 1$ an individual factor is undefined. Every right half-plane contains a positive real point, so no right half-plane supports the required finite nonzero ordinary product. □

Remark 5 (Exact scope of the theorem). Theorem 4 rules out the *ordinary orbit-counting* product (4). Replacing all orbits in Γ_p by one factor changes the primitive counting measure from one atom per orbit to one atom per packet. Such a measured, clean-family, groupoid, or cohomological collapse could be meaningful after a new theorem supplies its coefficient. The theorem above neither constructs nor excludes every such enrichment.

5 Four independent trace-normalization gates

The packet theorem and the ordinary-product obstruction leave open a more structured replacement. Before one can speak of a packet determinant, however, four interfaces must be distinguished. Figure 1 shows the logical order. Possessing an earlier object does not automatically define the next.

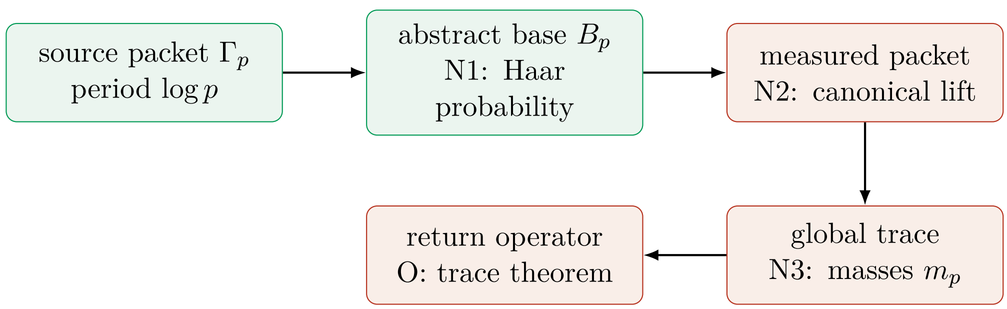

Figure 1: The packet-to-determinant interface. Green boxes are available from the source or an abstract compact-group theorem. Red boxes are not supplied for the frozen object. N1 alone is not a packet trace.

5.1 N1: normalized Haar probability on the abstract base

Given the compact Hausdorff group B_p, the Haar theorem gives a unique translation-invariant probability measure μ_p with $\mu_p(B_p) = 1$. This statement is PROVED as an abstract compact-group fact. Deninger does not define μ_p in the packet theorem.

The word canonical must be qualified. Normalized Haar probability is canonical *after the abstract group B_p and its topology have been fixed.* The source's projection from packet coordinates to B_p depends on a lift and an embedding. The transition maps between different choices are not classified, so the source does not prove that transporting μ_p to the quotient orbit set is choice-independent. That source-to-packet assertion remains OPEN.

5.2 N2: lift or disintegration to the packet

Even a canonical measure on an orbit base does not uniquely specify a measure or a periodic distribution on the total packet. A lift would need at least:

1. a canonical measurable quotient map from Γ_p;
2. conditional measures along the circular flow fibres;
3. measurable or continuous variation of those conditional measures;
4. compatibility with isotropy and repeated coverings;
5. flow or holonomy invariance; and
6. independence of the auxiliary coordinates in (3).

Each individual orbit, viewed abstractly as the homogeneous circle $\mathbb{R}/(\log p)\mathbb{Z}$, does have a unique normalized invariant Haar probability. What is missing is a source-canonical measurable assembly of those probabilities over the packet base, independence from the auxiliary packet coordinates, and a theorem identifying fibre probability with a return-trace coefficient. The period $\log p$ also gives a natural length measure of total mass $\log p$; neither that measure nor normalized probability is automatically the trace-derived choice. No disintegration and trace theorem making these distinctions is supplied. N2 is therefore OPEN at the conceptual level and NOT TESTABLE as a frozen-source construction.

5.3 N3: relative masses across prime components

Suppose, conditionally, that every packet has acquired a normalized local functional. This still does not select the mass of the whole packet relative to the other prime components. The following elementary theorem isolates the remaining freedom without pretending that its algebra is already present in the source.

Theorem 6 (Conditional component-mass non-uniqueness). *Assume an algebraic direct sum of $*$-algebras, with componentwise product and involution,*

$$\mathcal{A}_{\mathrm{alg}} = \bigoplus_p \mathcal{A}_p,$$

where every $\mathcal{A}_p$ carries a nonzero positive trace τ_p. Assume also a component-preserving action by $*$-automorphisms whose restriction to each component leaves τ_p invariant. Then every positive sequence $m = (m_p)_p$ defines

$$\tau_m((a_p)_p) = \sum_p m_p \tau_p(a_p), \tag{5}$$

and τ_m is positive, tracial, additive, and invariant under the component-preserving action. Componentwise homogeneous axioms therefore do not determine the relative masses m_p.

Proof. Every element of $\mathcal{A}_{\mathrm{alg}}$ has finite support, so (5) is well defined. Linearity is immediate. Positivity follows from $m_p > 0$ and positivity of each τ_p. For $a, b \in \mathcal{A}_{\mathrm{alg}}$,

$$\tau_m(ab) = \sum_p m_p \tau_p(a_p b_p) = \sum_p m_p \tau_p(b_p a_p) = \tau_m(ba).$$

The assumed local invariance and direct-sum additivity give the corresponding properties componentwise. If two sequences differ at an index where $\tau_p \neq 0$, the corresponding functionals are distinct. □

The hypotheses in Theorem 6 are substantive. The frozen source provides neither the algebras $\mathcal{A}_p$, their traces and domains, nor the absence of cross-prime relations. On a completed algebra, boundedness, lower semicontinuity, semifiniteness, or summability can restrict the allowable sequences. A future arithmetic category could also impose relations between components. Thus the theorem is CONDITIONAL, while the existence and uniqueness of the source-intrinsic trace remain NOT TESTABLE.

A still simpler version is global rescaling: if a nonzero trace τ satisfies only homogeneous axioms, then $c\tau$ satisfies the same axioms for every $c > 0$. A nonhomogeneous normalization such as $\tau(1) = 1$ would break this freedom, but the relevant algebra, unit, and source-derived normalization must first be specified.

5.4 O: operator or fixed-point trace

N1–N3 would not by themselves produce a dynamical trace formula. One still needs a Hilbert, Banach, cohomological, or distributional domain; a return or propagation operator; a trace-class, flat-trace, or regularized-trace criterion; and a theorem deriving the periodic coefficient. For example, under any full-support realization in which the return at time $\log p$ acts as the identity on an infinite-dimensional $L^2(\Gamma_p)$, the ordinary Hilbert-space trace is unavailable because the identity is not trace class. This conditional observation does not exclude other flat or cohomological traces; it shows why a probability measure alone is not the operator theorem.

5.5 Copied-packet and arbitrary-base controls

Two controls expose hidden assumptions. First, if a geometric component P is replaced by a disjoint union $P \sqcup P$, trace additivity requires

$$\tau(P \sqcup P) = 2\tau(P).$$

Renormalizing both P and $P \sqcup P$ to total mass one would violate additivity; counting once per isomorphism class would erase geometric multiplicity. This is a consistency control, not an existence no-go.

Second, any infinite compact group can be used as the base of a collection of equal-period circles and has normalized Haar probability. Haar probability alone therefore proves too much: it is not arithmetic-specific. A successful packet trace must use source-derived arithmetic or Frobenius structure at a clearly identified trace step.

6 Formal packet products and the abscissa diagnostic

To measure what is left undetermined, suppose only for this section that packet masses m_p have been supplied externally. The formal packet product and its logarithmic derivative would be

$$Z_m(s) = \prod_p (1 - p^{-s})^{-m_p}, \tag{6}$$

$$\log Z_m(s) = \sum_p \sum_{r \geq 1} \frac{m_p}{r} p^{-rs}, \tag{7}$$

$$-\frac{Z_m'}{Z_m}(s) = \sum_p \sum_{r \geq 1} m_p (\log p) p^{-rs}. \tag{8}$$

For nonintegral m_p, the product notation means the exponential of (7) on its half-plane of absolute convergence. Equivalently, for $\Re s > 0$ it uses the holomorphic power-series branch $-\operatorname{Log}(1-p^{-s}) = \sum_{r\geq 1} p^{-rs}/r$. No continuation is implied by this convention. These equations are a sensitivity model, not a determinant constructed from the flow. In particular, choosing $m_p = 1$ gives the desired one-packet factor and the familiar repetition coefficient by stipulation. The coefficient cannot then be cited as evidence that the stipulation was dynamically derived.

Theorem 7 (Packet mass determines the absolute abscissa)**.** *Let $m_p > 0$, let $\alpha \geq -1$, and suppose that for all sufficiently large primes*

$$cp^{\alpha} \leq m_p \leq Cp^{\alpha}$$

for constants $c, C > 0$. Then the positive series

$$\sum_p \sum_{r\geq 1} \frac{m_p}{r} p^{-r\sigma}$$

converges exactly when $\sigma > \alpha + 1$. Thus the formal packet product's absolute-convergence abscissa is $\alpha + 1$, and is not determined by the period list alone.

Proof. If $\sigma \leq 0$, the $r = 1$ subseries has terms comparable to $p^{\alpha-\sigma}$, whose exponent is at least -1. It diverges by Euler's reciprocal-prime theorem at exponent -1, and by termwise domination above that exponent. Hence it remains to consider $\sigma > 0$. For $\sigma > 0$,

$$\sum_{r\geq 1} \frac{p^{-r\sigma}}{r} = -\log(1 - p^{-\sigma}).$$

For all sufficiently large p, this quantity is bounded above and below by positive constant multiples of $p^{-\sigma}$. The double series therefore converges if and only if

$$\sum_p m_p p^{-\sigma} \asymp \sum_p p^{-(\sigma-\alpha)}$$

converges. The prime subseries converges for $\sigma - \alpha > 1$, by comparison with the corresponding integer p-series, and diverges for $\sigma - \alpha \leq 1$: at equality this is Euler's divergent reciprocal-prime series, while below equality its positive terms dominate $1/p$ for large p. Since $\alpha \geq -1$, the claimed domain lies in $\sigma > 0$, where the logarithmic expansion is valid. □

Corollary 8. *Unit masses have formal abscissa 1; masses $p^{-1/2}$, p, and p^{-1} have formal abscissae 1/2, 2, and 0, respectively. Consequently a request that the product converge for $\Re s > 1$ cannot be used to infer unit packet mass: that would select the normalization from the desired output.*

Signed masses do not remove the provenance problem. Conditional cancellations may alter a signed finite sum while leaving the absolute first-return majorant

$$\sum_p |m_p| p^{-\sigma}$$

unchanged. A phase or sign must be produced by the same return trace, not chosen to improve convergence or match a target.

7 Zero-free computational controls

7.1 Design and leakage boundary

The accompanying script generates rational primes deterministically only after the packet-weight ambiguity and Theorem 7 have been stated. Prime generation here evaluates closed-point-indexed formulas; it is not used to build the candidate dynamics. The script imports no Riemann-zero table, calls no zeta-zero routine, fits no scale, and fits no packet mass.

Four artifacts are produced:

1. finite-orbit partial sums within fixed equal-period packets;
2. packet-product sensitivity for five disclosed mass models;
3. a repetition ledger for the first ten primes and five repetitions; and
4. a copied-packet additivity control.

The main run used the 9,592 primes at most 100,000, five test values of σ, and the mass models 1, $p^{-1/2}$, p, p^{-1}, and a modulo-four sign. Five unit tests passed. A manifest records the cutoff, forbidden inputs, interpretation boundary, and SHA-256 hash of every result file.

7.2 Multiplicity and copied-packet results

For N disclosed individual orbits of period $\log p$, the finite log product is exactly

$$L_{p,\sigma}(N) = -N \log(1 - p^{-\sigma}). \tag{9}$$

At $p = 2$ and $\sigma = 1.25$, it increases from 0.545500245954 for $N = 1$ to 69.8240314821 for $N = 128$. This linear finite calculation illustrates the net used in Theorem 4; it does not prove uncountability, which is supplied by Proposition 1.

The copied-packet file verifies for one, two, and three copies that log contributions scale exactly by the copy number. For example, at the same $(p, \sigma) = (2, 1.25)$, the values are 0.545500245954, 1.09100049191, and 1.63650073786. The control detects the inconsistency in silently renormalizing a disjoint union back to mass one.

7.3 Packet-mass sensitivity

Table 2 reports selected finite log products. The numbers are not convergence proofs; their role is to make the theorem's normalization dependence reproducible.

Table 2: Selected values of $\sum_{p \le X} -m_p \log(1 - p^{-\sigma})$.

Mass m_p	σ	predicted abscissa	$X = 100$	$X = 100{,}000$
1	0.75	1	3.613734	10.395150
p^{-1}	0.75	0	0.816878	0.824218
1	1.25	1	1.356372	1.509763
$p^{-1/2}$	1.25	0.5	0.714463	0.721743
p	1.25	2	11.761876	751.614398

For the modulo-four sign model at $\sigma = 0.75$, the signed finite log product changes only from -0.581723 to -0.658170, while the absolute first-return mass grows from 2.414362 to 9.179910. This is a direct warning against mistaking cancellation for a source-derived trace phase.

All computational results are finite deterministic illustrations. They are not existence tests for a trace, do not establish that arbitrary weights are admissible on an unspecified completed algebra, and do not carry any Route-A verdict independently of the proofs.

8 Why established trace frameworks do not fill the gap

Positive-dimensional families of periodic points can contribute to rigorous trace formulas. The obstruction is not the word family; it is the absence of the geometry, density, and operator hypotheses that make the family traceable. This section audits five relevant interfaces.

8.1 Clean Hamiltonian and wave traces

In the Duistermaat–Guillemin setting, a positive elliptic pseudodifferential operator on a closed smooth manifold generates a Hamiltonian bicharacteristic flow on a smooth symplectic cotangent bundle. At a period T, the fixed set may have positive dimension, but it must be clean:

$$T_z \operatorname{Fix}(\Phi^T) = \ker(d\Phi_z^T - I). \tag{10}$$

The trace singularity is then obtained by integrating a canonical density on each clean component. That density uses the symplectic normal linearization, the operator's symbol and half-density data, and Maslov information; it is not the uniform rule that every component counts once [6].

For Γ_p, the frozen source gives no finite-dimensional smooth atlas, derivative $d\phi^T$, smooth fixed submanifold, symplectic normal bundle, principal symbol, or Maslov data. Thus even the clean identity (10) cannot be stated on the current object. This proves inapplicability of the cited theorem, not impossibility of constructing a future smooth enrichment.

8.2 Contact Morse–Bott families

The Morse–Bott contact condition likewise permits nonisolated periodic families only after substantial structure has been fixed. For a compact smooth contact manifold with contact form α, the period-T set N_T must be a closed smooth submanifold, $d\alpha|_{N_T}$ must have locally constant rank, and

$$T_x N_T = \ker(d\phi_x^T - I).$$

Bourgeois uses these conditions to construct a Morse–Bott approach to contact homology [2]; the work itself is not a general packet trace formula. The rational-Witt packet presently has neither the contact manifold nor the differential return data needed to test these conditions.

8.3 Foliated and transversally elliptic traces

Kordyukov's relative Duistermaat–Guillemin theorem starts with a compact smooth foliated manifold and a positive self-adjoint transversally elliptic pseudodifferential operator with holonomy-invariant transverse principal symbol. The relative fixed locus must satisfy a smooth clean fibre-product condition, and a mollified propagator is made trace class before its trace singularities are analyzed [11]. A transverse probability by itself supplies none of these operator inputs.

The recent trace formula for foliated flows of Álvarez López et al. [1] is particularly relevant because it is motivated by Deninger-type formulas. Its hypotheses nonetheless remain specific: a closed smooth manifold, a smooth transversely oriented codimension-one foliation, a foliated smooth flow, simple closed orbits, and transversely simple preserved leaves, together with leafwise currents, smoothing b-pseudodifferential operators, and a regularized b-trace. The simple closed-orbit condition excludes the normal eigenvalue 1, so those closed orbits are locally isolated. Positive-dimensional geometric terms in that formula are preserved leaves, not compact families of closed orbits. Moreover, the b-trace is regularized and need not vanish on every commutator; some zero-time terms depend on auxiliary choices. It cannot be replaced by packet-base Haar probability.

8.4 Groupoid Haar systems, transverse measures, and traces

Three constructions are often compressed into the word trace:

1. A Haar system on a second-countable locally compact groupoid is a continuous invariant family of Radon measures on range fibres. It permits convolution on $C_c(G)$. Existence and uniqueness are not automatic [12].
2. A holonomy-invariant transverse measure is additional measured data. Under the appropriate foliation hypotheses it induces a trace or dimension map on a foliation operator algebra [3]. It also produces a Ruelle–Sullivan current [14].
3. A dynamical fixed-point trace requires a specified return, propagation, transfer, or cohomological operator and a theorem identifying its trace distribution with periodic geometry.

The implications do not run automatically from the first item to the second or third. A longitudinal Haar system for a standard transformation groupoid of the $\mathbb{R}$-flow, for example, would not choose a transverse mass on the uncountable orbit base or relative masses across primes.

Several transformation or equivalence-relation groupoids could be built externally from the source-defined $\mathbb{Q}_{>0}$ and $\mathbb{R}$ actions. The source does not select one as the periodic trace groupoid, define its arrow topology for this purpose, choose a Haar system and transverse measure, or specify an observable completion. Freezing any such choice would create a new versioned candidate, not reveal a trace already present in `DEN-WITT-Z-FIN`.

8.5 Isolated Anosov traces as a coefficient benchmark

For compact smooth Anosov flows, the microlocal flat trace is defined only after a wavefront-set condition permits restriction of the propagator kernel to the diagonal. The resulting Guillemin identity has the form

$$\operatorname{tr}^\flat e^{-itP} = \sum_\gamma \frac{T_\gamma^\# \, \delta(t - T_\gamma)}{|\det(I - \mathcal{P}_\gamma)|}, \qquad t > 0, \tag{11}$$

where $T_\gamma^\#$ is the primitive period and $\mathcal{P}_\gamma$ is the normal Poincaré return map [7]. Hyperbolicity makes the normal determinant nonzero. Alternating exterior-power traces then lead to a Ruelle zeta function.

Ruelle's original Fredholm construction assumes analytic hyperbolic structure [13]. Later work using anisotropic spaces and microlocal methods gives meromorphic continuation in smoother Anosov settings [7, 9]. The 2022 arXiv erratum to the former corrects part of its contact spectral-gap analysis while stating that its meromorphic-continuation result is unaffected [10]. These theorems are rigorous benchmarks for how primitive period, repetition, normal stability, and an operator trace fit together. They concern isolated hyperbolic periodic orbits and do not assign mass one to a topological packet.

For the compact hyperbolic/Fuchsian benchmark in the cited setting, the direct-product convention also yields the exact identity

$$R_\Gamma(s) = \prod_{\gamma_0}(1 - N_{\gamma_0}^{-s}) = \frac{Z_\Gamma(s)}{Z_\Gamma(s+1)}$$

and therefore

$$\frac{R'_\Gamma}{R_\Gamma}(s) = \sum_{\gamma_0}\sum_{r\geq 1}(\log N_{\gamma_0})N_{\gamma_0}^{-rs}$$

up to the displayed logarithmic-derivative convention [8]. This is an exact isolated-orbit coefficient calibration. It does not specialize to Deninger's packet system because the orbit cardinality, smooth normal return, and trace domain are different.

Table 3: Applicability matrix for positive-dimensional or measured trace frameworks.

Framework	Geometric gate	Analytic or measured gate	Frozen packet status
clean Hamiltonian/PDO	smooth symplectic flow, clean fixed submanifold, normal linearization	elliptic propagator, symbols, canonical density, Maslov data	geometry and operator absent
contact Morse–Bott	compact contact manifold, smooth N_T, tangent identity, constant-rank $d\alpha$	contact-homology package; not itself a trace formula	contact and return data absent
transversally elliptic	compact smooth foliation and clean relative fixed set	positive self-adjoint transversally elliptic operator and mollified trace	manifold, operator, and clean test absent
foliated-flow b-trace	smooth codimension-one foliation; simple orbits and special preserved leaves	leafwise currents, smoothing b-PDO, regularized b-trace	packet family is neither input type
locally compact groupoid	selected arrow topology and range fibres	Haar system gives convolution; transverse trace is extra data	no packet trace groupoid selected
Anosov flat trace	smooth hyperbolic flow and isolated normal-nondegenerate orbits	wavefront-admissible flat trace and transfer generator	uncountable equal-period packets violate the orbit model

The table supports an applicability verdict: existing theorems cannot be invoked on the frozen packet object because their hypotheses are absent or not testable. It does not prove that a new packet-specific theorem cannot be developed.

9 Deninger Section 11: a genuine algebra on a different object

It would be inaccurate to say that the primary source contains no algebra or Haar normalization. Section 11 constructs both. The relevant object is a locally compact zero-dimensional inverse-limit group, denoted $\overleftarrow{K}^\times$, with a compact open subgroup. Haar measure is normalized by assigning that subgroup mass one, convolution is defined on $C_c(\overleftarrow{K}^\times)$, and a source-defined submonoid gives a nonunital convolution subalgebra. The section further constructs a Galois-equivariant algebra homomorphism from that subalgebra to continuous functions on the rational-Witt point space [5].

All of these are PROVED source structures. Their presence makes the missing bridge more precise. The inverse-limit group and its convolution algebra are not identified with B_p, $\Gamma_p/\mathbb{R}$, or a holonomy groupoid of the packet. The algebra homomorphism is not a trace functional. Section 11 does not provide:

1. a map from its group convolution to the packet orbit relation;
2. a representation of the suspension flow on that convolution algebra;
3. a periodic-return kernel or operator;
4. a trace, flat trace, or semifinite weight whose fixed-time support is Γ_p;
5. a formula for primitive and repeated packet coefficients; or
6. a Fredholm or nuclear determinant theorem.

Consequently the normalization of Haar measure on $\overleftarrow{K}^\times$ fixes the scale of that *group convolution*. It does not fix the N2 packet lift, N3 relative packet masses, or O-gate return trace in Section 5.

Table 4: Section-11 bridge test.

Required map or theorem	Frozen-source status
inverse-limit locally compact group and normalized Haar convolution	PROVED
algebra homomorphism into functions on the rational-Witt point space	PROVED
map to $\Gamma_p/\mathbb{R}$ or packet groupoid	NOT TESTABLE
flow-return representation and trace domain	NOT TESTABLE
source-derived cross-packet masses and repetition theorem	NOT TESTABLE
Fredholm determinant or divisor theorem	NOT TESTABLE

The overview article also recalls a distributional transverse-index formula for a different class of smooth compact foliated systems [4]. That formula assumes smoothness, ellipticity, transversality, and related analytic data, and uses a mollified trace because fixed-time operators need not be trace class. The rational-Witt spaces are later described as infinite-dimensional and not yet satisfactorily understood. The overview supplies motivation, not a transfer theorem proving that the smooth trace formula applies to Γ_p.

10 Route-A evaluation

The gate decision must preserve the split between a disproved ordinary construction and an undefined enriched one. Table 5 records that distinction.

The full Route-A calibration is `A0_ANALYTIC_ARITHMETIC_ORIGIN`, `A1_WEAK`, and `A2_FAIL`; the frozen-source measured alternative at A2 is `NOT_TESTABLE`. The registry records `A3_FAIL`, but that gate is substantively not reached because A2 supplies no candidate determinant. Finally,

Table 5: A2 split verdict for the same frozen periodic object.

Candidate construction	What is mathematically available?	Verdict
ordinary product over every primitive orbit	finite-subset net, divergent by Theorem 4	`A2_FAIL` (PROVED)
packet product with $m_p = 1$	formal expression after imposing the normalization	`A2_FAIL` for dynamical provenance
homogeneous direct-sum trace	conditional family τ_m; normalization remains free	`A2_FAIL` for canonicality under the stated axioms
trace intrinsic to the frozen source	no selected algebra, domain, lift, global mass, or return theorem	`NOT_TESTABLE`
Section-11 convolution as packet trace	convolution algebra exists; packet-return bridge does not	`NOT_TESTABLE`

`A4_FAIL` records that no quantum or cohomological lift is constructed. The overall status remains `ROUTE_A_EXPLORATORY`. The A3 serialization is a downstream gate failure, not evidence that an existing determinant failed continuation or root matching. Continuation domain, divisor, functional equation, cutoff drift, missing roots, and extra roots all have the registry reason `not_applicable_no_candidate_determinant`.

The arithmetic gate passes because the packet index and clock are derived from closed points. A1 is weak because the periodic structure is genuine but does not have a canonical isolated-orbit multiplicity, normal monodromy, phase, or trace weight. A2 has no pass for the reasons proved and delimited above. No later layer can repair an undefined determinant retroactively. Accordingly Route B is not authorized by this paper.

10.1 Smallest positive next theorem

The minimum non-circular advance is not another formal Euler product. It is a versioned enrichment supplying all of:

1. a choice-independent topological or measured orbit quotient, or a source-natural packet groupoid;
2. a compatible Haar system or transverse measure, including a proof of invariance under all coordinate changes;
3. a canonical lift or disintegration to the packet;
4. a global arithmetic rule for component masses;
5. an observable, return, or cohomological operator with a legitimate trace domain; and
6. a fixed-point or Lefschetz theorem deriving primitive and repeated coefficients from that same structure.

The expected Euler factor, convergence half-plane, or Riemann divisor may be used only after those inputs are fixed. Otherwise the target would choose the normalization under test.

11 Limitations

This work is an exact audit of one versioned candidate, not a universal no-flow theorem.

First, Proposition 1 uses the source's noncanonical set-level identification of the orbit set. It proves cardinality and length-local nonfiniteness, but not that the packet quotient has the product topology of B_p, that the orbit family is smooth or continuous, or that the packet is locally trivial.

Second, the finite-kernel admissibility class is a disclosed source-permitted choice, not a uniquely selected arithmetic condition. A different admissibility class changes the frozen candidate and requires a new evaluation. The current no-go applies to the ordinary orbitwise product of the selected full-packet system.

Third, Theorem 6 is conditional on an algebraic direct sum, local traces, component preservation, and only homogeneous componentwise axioms. It does not assert that every positive sequence extends to a bounded or semifinite trace on every completion. Nor does it exclude a future arithmetic functor that relates the packet components and selects their masses.

Fourth, failure of (4) is not failure of every measured, clean, groupoid, distributional, or cohomological determinant. Those alternatives are classified as open or not testable because the current source does not supply their input data. This evidential distinction is the main scope boundary of the paper.

Fifth, the computational audit is finite. Its multiplicity and copied-packet relations are exact within the generated rows, while its cutoff tables merely illustrate the analytic theorem. It provides no independent evidence for a trace and uses no Riemann-zero data.

Finally, the literature investigation was targeted to primary sources that can decide the packet-to-trace interface: clean and Morse–Bott fixed sets, foliated traces, groupoid Haar systems, transverse measures, and Anosov flat traces. It was not a systematic review of all noncommutative-geometric approaches. The local primary PDFs were checked with standard PDF metadata and text extraction. A reader-library preflight was unavailable in the environment, so page locators in the audit notes are advisory and were cross-checked against printed numbering rather than treated as a complete PDF-object certification.

12 Conclusion

Deninger's rational-Witt flow crosses a difficult arithmetic threshold: for $\operatorname{Spec}\mathbb{Z}$, rational primes and periods $\log p$ arise from the source geometry rather than from fitting. The same theorem reveals why an Euler product does not follow automatically. A prime indexes an uncountable packet of primitive equal-period orbits, not one isolated orbit. The ordinary orbitwise Ruelle product therefore diverges within one packet at every positive real parameter.

Collapsing a packet to one factor is a different construction. Normalized Haar probability exists on the abstract compact base, but the source has not proved a choice-independent packet measure, a global trace normalization, or an operator fixed-point coefficient. Conditional homogeneous trace axioms leave arbitrary component masses, and those masses move the formal convergence abscissa. Section 11 provides a real Haar-normalized convolution algebra, but on another inverse-limit group and without a packet-return bridge. Existing clean, foliated, groupoid, and microlocal trace theorems make clear what additional data are needed; none licenses the missing step on the frozen object.

The rigorous endpoint is consequently split. The conventional one-factor-per-orbit determinant fails. A packetwise measured or cohomological determinant remains open and presently not testable. Route A retains an exploratory arithmetic survivor, but no determinant reaches the analytic or quantum gates. The next genuine advance must construct the packet-to-trace interface itself, rather than infer it from the desired Euler factor.

Reproducibility and declarations

Data and code availability. All research protocols, source-audit notes, proof audit, deterministic Python code, tests, experiment script, result tables, checksums, figure source, and manuscript source are included in the `papers/2-flow-zeta/` project directory. From its `experiments/` subdirectory, the control suite is reproduced with

```
bash reproduce.sh
```

No proprietary dataset or Riemann-zero dataset is used.

Ethics statement. This theoretical and computational study involves no human participants, animals, intervention, personal data, or identifiable information. No institutional ethics review was required for the reported work.

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A Claim-to-evidence ledger

Table 6: Status of the manuscript's central claims.

Claim	Evidence label	Basis
packets exhaust the periodic set and have least period $\log p$	PROVED	Deninger Theorems 5.2 and 6.1
packet orbit set is noncanonically indexed by B_p	PROVED	source equations (37)–(39)
B_p is uncountable	PROVED	Proposition 1
packet quotient is canonically homeomorphic to B_p	OPEN	no source topology/transition theorem
ordinary individual-orbit product diverges	PROVED	Theorem 4
normalized Haar probability exists on abstract B_p	PROVED	compact-group Haar theorem

Claim	Evidence label	Basis
that probability canonically lifts to Γ_p	OPEN/NOT TESTABLE	quotient map and disintegration not source-defined
homogeneous component axioms determine m_p	CONDITIONAL	they do not under Theorem 6's stipulated hypotheses
intrinsic packet trace exists on the frozen object	NOT TESTABLE	algebra, domain, and return theorem absent
Section 11 is a Haar-normalized convolution algebra	PROVED	primary-source construction
Section 11 supplies a packet return trace	NOT TESTABLE	no source bridge
all enriched packet traces are impossible	not claimed	outside proved boundary

B Artifact map

Relative path	Purpose
`notes/research_protocol.md`	frozen question, validity criteria, ten proof obligations, and ten adversarial controls
`notes/devils_advocate_checkpoint1.md`	independent design-stage leakage and falsifiability audit
`notes/phase2_deninger_source_audit.md`	theorem-numbered reconstruction of the source packet and Section 11
`notes/phase2_trace_bibliography.md`	primary-source trace-framework applicability matrix and search ledger
`notes/phase3_trace_no_go_audit.md`	mathematical no-go boundary and three-layer normalization audit
`notes/proof_audit.md`	independent proof correction, status calibration, and result-hash verification
`code/packet_trace_controls.py`	deterministic multiplicity, weight, repetition, and copy controls
`code/test_packet_trace_controls.py`	five unit tests
`experiments/reproduce.sh`	one-command reproduction entry point
`results/orbit_multiplicity_divergence.csv`	finite equal-period multiplicity illustration
`results/packet_weight_sensitivity.csv`	disclosed packet-mass and cutoff sensitivity table
`results/packet_repetition_ledger.csv`	formal repeated-return terms under three mass models
`results/copied_packet_control.csv`	disjoint-copy additivity control
`results/packet_trace_controls_manifest.json`	run parameters, forbidden inputs, boundaries, and SHA-256 hashes
`paper/figures/trace_gate_diagram.tex`	N1–N3–O logical interface diagram

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